EMMANUELLA ESSIFUA ACHEAMPONG

Doctoral Researcher | Social Policy | Artificial Intelligence | Cybersecurity | Data Science

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ACADEMIC PROFILE

Interdisciplinary PhD student in Social Policy at Brandeis University, concentrating in Economic and Racial Equity. Research spans educational pathway disruptions, educational inequality and economic opportunity, alongside machine-learning-based fraud detection, bank-related identity theft, cybersecurity decision support and digital trust. Background in business analytics, banking, data science and cybersecurity.

EDUCATION

Brandeis University — Heller School for Social Policy and Management | PhD in Social Policy *2026–present*

- Concentration: Economic and Racial Equity; academic advisor: Joel Gershenfeld.

Brandeis University — International Business School | MS in Business Analytics (STEM) *2025*

- Graduated with distinction; Beta Gamma Sigma.

University of Professional Studies, Accra | BSc in Banking and Finance *2019*

- GPA: 3.90; Overall Best Student; Best Student in Banking and Finance.

RESEARCH EXPERIENCE & PROJECTS

Research Assistant — Center for Global Development and Sustainability, Brandeis University *2026–present*

- Cyber Innovation & Livelihood Security division.

Educational Pathway Disruptions in Ghana *2026–present · Ongoing*

- Examines school participation, persistence, transitions, completion and re-entry, with attention to inequality, institutions and policy implications.

Bank-Related Identity Theft in the U.S. *Manuscript in preparation*

- Examines bank-related identity theft and associated financial loss.

Human-in-the-Loop Cybersecurity Decision Framework *Ongoing research*

- Explores integration of machine-learning risk indicators with human review, decision escalation and accountability.

FraudLens™ *Research prototype*

- Developed experimental fraud classification models using IEEE-CIS data; compared XGBoost and Random Forest, performed feature engineering and model evaluation, and built a Streamlit demonstration.

Education, Learner Variability and Responsive School Design *Research exploration*

- Examines how learning differences and identity development can inform more responsive school practices; scope under development.

PROFESSIONAL EXPERIENCE

Consolidated Bank Ghana — Data Scientist / Data Analyst / Management Trainee *Nov 2021–Aug 2023*

- Contributed to SME eligibility analytics associated with approximately 20% less processing time.
- Developed forecasting approaches associated with approximately 60% shorter concession forecasting cycles.
- Built Power BI reporting and applied Python, R and data transformation tools across banking operations.

Miracle Cyber — Cybersecurity Analyst *May–Aug 2024*

- Investigated phishing, ransomware and data-loss-prevention scenarios using Microsoft Defender, AnyRun and AbuseIPDB.
- Contributed to cybersecurity awareness activities associated with a reported 25% improvement.

SELECTED SCHOLARLY ENGAGEMENTS

- Grace Hopper Celebration — Accepted speaker (2025)
- INFORMS IRAC Student Research Showcase — Winner (2025)
- Brandeis Global Cybersecurity & AI Conference — Presentation on bank-related identity theft
- Joint Statistical Meetings — Birds of a Feather (2025)

HONORS & FELLOWSHIPS

- 2026–2027 Sillerman Center Named Fellow, Brandeis University
- 2025 INFORMS IRAC Student Research Showcase — Winner
- 2025 Beta Gamma Sigma, Brandeis International Business School
- 2025 Master's Degree with Distinction, Brandeis University
- 2019 Overall Best Student, University of Professional Studies, Accra
- 2019 Best Student in Banking and Finance, University of Professional Studies, Accra
- Brandeis International Business School — Dean Scholarship and Peter Petri Global Fellowship

ACADEMIC SERVICE & LEADERSHIP

- Reviewer — HICSS 2026
- Reviewer — IEEE MIT Undergraduate Research Technology Conference 2025
- Reviewer — Grace Hopper Celebration (2025)
- Co-Chair — African Forum, Brandeis University (2026–present)

RESEARCH & TECHNICAL SKILLS

Quantitative research, survey analysis, regression, econometrics, literature reviews, policy analysis; Python, R, SQL, Power BI, Tableau, Excel/VBA, SPSS, scikit-learn, XGBoost, Random Forest, TensorFlow, Streamlit, Flask, BigQuery, Looker, AWS EC2, Git; Microsoft Defender, AnyRun, AbuseIPDB.

PROFESSIONAL STUDIES & LANGUAGES

Chartered Institute of Bankers, Ghana (2019–2022). Languages: English (fluent), Fante (native), Ga (native), French (beginner).